

Apogee: Two-Stage Ascent to Circular LEO (Equatorial Launch, No Earth Rotation)

Abstract

We define and derive, from first principles, a planar (2D) rocket ascent simulator to circular low Earth orbit (LEO) aligned to publicly available Falcon 9 data. The only mission input is target circular-orbit altitude. Earth rotation is neglected by design; launch is assumed on the equator (Ecuador latitude $\approx 0^\circ$) to keep the orbit equatorial while preserving the no-rotation requirement. The model includes variable mass, non-constant gravity, U.S. Standard Atmosphere 1976 density, and aerodynamic drag. Staging, pitch-over, and cutoff are handled as events. We provide the final numbered ODE system, event logic, orbital validation, and a strict numerical methodology (adaptive Runge–Kutta + root-finding shooting).

1 Project Definition (What We Solve)

1.1 Mission input

The single mission input is the desired circular orbit altitude h_{target} above mean sea level. Define

$$r_{\text{target}} = R_E + h_{\text{target}}. \quad (1)$$

1.2 Outputs

The simulator must return the full time history (trajectory) until orbit insertion:

$$t, r(t), h(t), \lambda(t), x(t), v(t), \gamma(t), m(t), q(t), M(t), D(t),$$

plus terminal orbit diagnostics at MECO/SECO: $(a, e, r_{\text{apo}}, r_{\text{peri}})$.

1.3 Design constraints (fixed by your requirements)

- **No Earth rotation** (inertial Earth is non-rotating).
- **Equatorial launch** (Ecuador line) interpreted as: the ascent plane is the equatorial plane and the target orbit inclination is 0° . With no Earth rotation, launch latitude does not modify inertial dynamics, but we keep geometry consistent.
- **Three guidance phases:** vertical ascent $\rightarrow$ pitch-over kick $\rightarrow$ gravity turn with thrust aligned to velocity ($\alpha = 0$).

2 Earth Constants (Required and Why)

These constants are required because gravity and orbital mechanics are central to the problem.

2.1 Constants

- Mean Earth radius: $R_E = 6\,371\,000\text{ m}$ (use as a geometric reference; any consistent Earth radius standard is acceptable).
- Earth gravitational parameter: $\mu = 3.986\,004\,418 \times 10^{14}\text{ m}^3/\text{s}^2$ (sets $g(r)$ and orbital speed).
- Standard gravity: $g_0 = 9.806\,65\text{ m/s}^2$ (definition used in I_{sp} relation).

2.2 Why these appear in the equations

Non-constant gravity:

$$g(r) = \frac{\mu}{r^2}. \quad (2)$$

Circular orbital speed at radius r :

$$v_{\text{circ}}(r) = \sqrt{\frac{\mu}{r}}. \quad (3)$$

3 Atmosphere Model (USSA76) and Using ussa1976

3.1 Why USSA76

We use the *U.S. Standard Atmosphere 1976* (USSA76) to obtain $\rho(h)$ and (optionally) $a_s(h)$ for Mach number. USSA76 is a published, engineering reference atmosphere used widely for trajectory and vehicle analyses [3].

3.2 Is ussa1976 OK to use?

Yes, as a practical implementation of USSA76: the PyPI package `ussa1976` provides a programmatic interface to the USSA76 model [4, 5]. This is appropriate for APOGEE because your requirement is a consistent atmosphere model, not real-time weather.

4 Falcon 9 Public (Official) Vehicle Inputs

4.1 Official Falcon 9 overview (SpaceX website)

SpaceX lists Falcon 9 height, diameter, gross mass, and payload-to-LEO capability [2]:

- Height: 70 m
- Core diameter: 3.7 m
- Vehicle mass (gross): 549 054 kg
- Payload to LEO: 22 800 kg (capability; **payload mass is parameterized** in APOGEE)

4.2 Official stage thrust and configuration (SpaceX User’s Guide)

SpaceX Falcon User’s Guide (2025) provides stage propulsion characteristics, including engine counts and stage thrust values [1]:

- Stage 1: 9 Merlin 1D (M1D), propellant LOX/RP-1; stage total thrust at sea level 7686 kN; engine thrust per engine up to 845 kN sea level; throttling capability noted [1].
- Stage 2: 1 Merlin Vacuum (MVac), propellant LOX/RP-1; stage thrust in vacuum 981 kN; throttling capability noted [1].

4.3 Derived geometry for drag reference area

Using diameter $d = 3.7$ m, define reference area

$$A_{\text{ref}} = \pi \left(\frac{d}{2} \right)^2 = \pi (1.85)^2 \approx 10.75 \text{ m}^2. \quad (4)$$

4.4 What SpaceX does *not* publish in these sources

The SpaceX website and the 2025 user guide do not consistently provide full stage-by-stage *dry mass*, *propellant mass*, or detailed I_{sp} curves. Therefore, if you want a numerically closed model *today*, you must choose one of:

1. treat stage masses and I_{sp} as **tunable engineering parameters**, calibrated to hit known staging and insertion performance; or
2. use secondary compilations (not “official SpaceX”), clearly labeled as assumptions.

APOGEE will proceed with option (1) by default (clean and honest).

5 Coordinate System and State

We use a planar, Earth-centered polar geometry (equatorial plane).

5.1 Definitions

- $r(t)$: geocentric radius
- $h(t) = r(t) - R_E$: altitude
- $\lambda(t)$: downrange central angle (rad)
- $x(t) = R_E \lambda(t)$: surface arc-length downrange (m)
- $v(t)$: speed magnitude
- $\gamma(t)$: flight-path angle measured from local horizontal (rad)

5.2 State vector

$$\mathbf{y}(t) = \begin{bmatrix} r \\ \lambda \\ v \\ \gamma \\ m \end{bmatrix}. \quad (5)$$

6 Kinematics: Step-by-Step Derivation

6.1 Velocity decomposition

In polar coordinates:

$$\mathbf{v} = \dot{r} \hat{\mathbf{e}}_r + r \dot{\lambda} \hat{\mathbf{e}}_\lambda. \quad (6)$$

Define γ such that

$$\dot{r} = v \sin \gamma, \quad r \dot{\lambda} = v \cos \gamma. \quad (7)$$

Hence the kinematic ODEs:

$$\boxed{\dot{r} = v \sin \gamma}, \quad \boxed{\dot{\lambda} = \frac{v \cos \gamma}{r}}, \quad \boxed{\dot{x} = R_E \dot{\lambda} = R_E \frac{v \cos \gamma}{r}}. \quad (8)$$

7 Forces and Models

7.1 Gravity (non-constant)

Central gravity acceleration:

$$\mathbf{a}_g = -\frac{\mu}{r^2} \hat{\mathbf{e}}_r. \quad (9)$$

7.2 Drag (USSA76 density)

Drag magnitude:

$$\boxed{D = \frac{1}{2} \rho(h) C_D(M) A_{\text{ref}} v^2}. \quad (10)$$

Mach number:

$$M = \frac{v}{a_s(h)}. \quad (11)$$

7.3 Thrust direction and angle of attack

Define unit tangent along velocity and unit normal:

$$\hat{\mathbf{t}} = \sin \gamma \hat{\mathbf{e}}_r + \cos \gamma \hat{\mathbf{e}}_\lambda, \quad \hat{\mathbf{n}} = \cos \gamma \hat{\mathbf{e}}_r - \sin \gamma \hat{\mathbf{e}}_\lambda. \quad (12)$$

Let α be angle of attack (AoA) between thrust and velocity direction:

$$\hat{\mathbf{u}}_T = \cos \alpha \hat{\mathbf{t}} + \sin \alpha \hat{\mathbf{n}}. \quad (13)$$

Then thrust acceleration:

$$\mathbf{a}_T = \frac{T}{m} \hat{\mathbf{u}}_T. \quad (14)$$

Drag acceleration is opposite velocity:

$$\mathbf{a}_D = -\frac{D}{m} \hat{\mathbf{t}}. \quad (15)$$

8 Variable Mass: Step-by-Step Derivation

Specific impulse definition:

$$I_{sp} = \frac{T}{\dot{m}_{\text{prop}} g_0}. \quad (16)$$

Vehicle mass decreases as $\dot{m} = -\dot{m}_{\text{prop}}$, so

$$I_{sp} = \frac{T}{(-\dot{m})g_0} \Rightarrow \boxed{\dot{m} = -\frac{T}{I_{sp}g_0}}.$$

9 Dynamics: Step-by-Step Derivation of $\dot{v}$ and $\dot{\gamma}$

Total acceleration:

$$\mathbf{a} = \mathbf{a}_T + \mathbf{a}_D + \mathbf{a}_g. \quad (17)$$

9.1 Tangential (speed) equation

By definition of speed along the velocity direction:

$$\dot{v} = \mathbf{a} \cdot \hat{\mathbf{t}}. \quad (18)$$

Compute each term:

$$\mathbf{a}_T \cdot \hat{\mathbf{t}} = \frac{T}{m} \hat{\mathbf{u}}_T \cdot \hat{\mathbf{t}} = \frac{T}{m} \cos \alpha, \quad \mathbf{a}_D \cdot \hat{\mathbf{t}} = -\frac{D}{m}, \quad \mathbf{a}_g \cdot \hat{\mathbf{t}} = \left(-\frac{\mu}{r^2} \hat{\mathbf{e}}_r\right) \cdot \hat{\mathbf{t}} = -\frac{\mu}{r^2} \sin \gamma.$$

Thus

$$\boxed{\dot{v} = \frac{T}{m} \cos \alpha - \frac{D}{m} - \frac{\mu}{r^2} \sin \gamma.} \quad (19)$$

9.2 Normal (turning) equation

A standard polar-kinematics identity yields

$$\mathbf{a} \cdot \hat{\mathbf{n}} = v\dot{\gamma} - \frac{v^2}{r} \cos \gamma. \quad (20)$$

Now compute $\mathbf{a} \cdot \hat{\mathbf{n}}$ from forces:

$$\mathbf{a}_T \cdot \hat{\mathbf{n}} = \frac{T}{m} \sin \alpha, \quad \mathbf{a}_D \cdot \hat{\mathbf{n}} = 0, \quad \mathbf{a}_g \cdot \hat{\mathbf{n}} = \left(-\frac{\mu}{r^2} \hat{\mathbf{e}}_r\right) \cdot \hat{\mathbf{n}} = -\frac{\mu}{r^2} \cos \gamma.$$

So

$$\frac{T}{m} \sin \alpha - \frac{\mu}{r^2} \cos \gamma = v\dot{\gamma} - \frac{v^2}{r} \cos \gamma.$$

Solve for $\dot{\gamma}$:

$$\boxed{\dot{\gamma} = \frac{T}{mv} \sin \alpha + \left(\frac{v}{r} - \frac{\mu}{r^2 v}\right) \cos \gamma.} \quad (21)$$

10 Gravity Turn Justification and Simplification

In the gravity-turn phase we enforce **thrust aligned with velocity**:

$$\boxed{\alpha = 0 \quad \Rightarrow \quad \mathbf{T} \parallel \mathbf{V}.} \quad (22)$$

Then the coupled equations simplify to

$$\boxed{\dot{v} = \frac{T}{m} - \frac{D}{m} - \frac{\mu}{r^2} \sin \gamma}, \quad \boxed{\dot{\gamma} = \left(\frac{v}{r} - \frac{\mu}{r^2 v} \right) \cos \gamma}. \quad (23)$$

This is physically consistent: thrust produces tangential acceleration, while gravity and curvature produce turning.

11 Final Numbered ODE System (Implementation-Ready)

Define closures:

$$h = r - R_E, \quad \rho = \rho(h) \text{ (USSA76)}, \quad a_s = a_s(h) \text{ (USSA76)}, \quad M = v/a_s,$$

$$D = \frac{1}{2} \rho(h) C_D(M) A_{\text{ref}} v^2.$$

Then

$$\boxed{\dot{r}} = \boxed{v \sin \gamma} \quad (24)$$

$$\boxed{\dot{\lambda}} = \boxed{\frac{v \cos \gamma}{r}} \quad (25)$$

$$\boxed{\dot{v}} = \boxed{\frac{T}{m} \cos \alpha - \frac{D}{m} - \frac{\mu}{r^2} \sin \gamma} \quad (26)$$

$$\boxed{\dot{\gamma}} = \boxed{\frac{T}{mv} \sin \alpha + \left(\frac{v}{r} - \frac{\mu}{r^2 v} \right) \cos \gamma} \quad (27)$$

$$\boxed{\dot{m}} = \boxed{-\frac{T}{I_{sp} g_0}} \quad (28)$$

During coast: set $T = 0$ so $\dot{m} = 0$.

12 Hybrid Events (Falcon 9–Aligned, Simplified)

12.1 Phase A: Vertical ascent

Hold $\gamma = \pi/2$ until a trigger (choose altitude h_{po}). Event:

$$g_{po}(\mathbf{y}) = (r - R_E) - h_{po} = 0.$$

Purpose: avoid the $1/v$ term near $v \rightarrow 0$ in (27).

12.2 Phase B: Pitch-over kick

At $t = t_{po}$ set (instantaneous reset):

$$\boxed{\gamma(t_{po}^+) = \frac{\pi}{2} - \theta_0.} \quad (29)$$

12.3 Phase C: Gravity turn

Set $\alpha = 0$ for the remainder of powered ascent.

12.4 Stage 1 $\rightarrow$ Stage 2 separation (event-driven)

We model a two-stage stack consistent with Falcon 9 being a two-stage LOX/RP-1 vehicle [1]. In the simplified model:

- Stage 1 burn ends when m reaches a stage-1 burnout mass $m_{1,\text{end}}$.
- Then drop stage-1 dry mass instantly and switch (T, I_{sp}, C_D, A) to stage-2 values.

Event (burnout):

$$g_{S1}(\mathbf{y}) = m - m_{1,\text{end}} = 0.$$

Reset:

$$m^+ = m^- - m_{1,\text{dry}}.$$

12.5 Final cutoff (MECO/SECO) chosen for convergence

To guarantee a solvable shooting problem, use a cutoff mass m_{cut} on the final stage:

$$g_{\text{cut}}(\mathbf{y}) = m - m_{\text{cut}} = 0, \quad T \leftarrow 0.$$

13 Orbit Validation (Two-Body Mechanics)

At cutoff, compute

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r}, \quad h_{\text{ang}} = rv \cos \gamma. \quad (30)$$

Eccentricity:

$$e = \sqrt{1 + \frac{2\varepsilon h_{\text{ang}}^2}{\mu^2}}. \quad (31)$$

Semi-major axis (bound orbit: $\varepsilon < 0$):

$$a = -\frac{\mu}{2\varepsilon}. \quad (32)$$

Apoapsis/periapsis radii:

$$r_{\text{apo}} = a(1 + e), \quad r_{\text{peri}} = a(1 - e). \quad (33)$$

14 Shooting Method (What We Solve Numerically)

14.1 Decision variables

We solve for

$$\mathbf{u} = \begin{bmatrix} \theta_0 \\ m_{\text{cut}} \end{bmatrix}.$$

14.2 Residual equations

We target circular insertion at r_{target} :

$$F_1(\mathbf{u}) = r_{\text{cut}}(\mathbf{u}) - r_{\text{target}}, \quad F_2(\mathbf{u}) = v_{\text{cut}}(\mathbf{u}) - \sqrt{\frac{\mu}{r_{\text{target}}}}. \quad (34)$$

Success diagnostics (not necessarily enforced as equalities at first):

$$|\gamma_{\text{cut}}| \leq \epsilon_\gamma, \quad |r_{\text{apo}} - r_{\text{target}}| \leq \epsilon_r, \quad |r_{\text{peri}} - r_{\text{target}}| \leq \epsilon_r.$$

15 Numerical Methods (Strict Recommendations)

15.1 ODE integration

Use adaptive embedded explicit Runge–Kutta (nonstiff):

- Dormand–Prince 5(4) (Dopri5), or Tsitouras 5(4) (Tsit5).

Use event detection (root finding on $g(\mathbf{y})$) for: pitch-over, stage separation, cutoff.

15.2 Nonlinear solve for shooting

Solve $\mathbf{F}(\mathbf{u}) = \mathbf{0}$ using:

- Powell hybrid / trust-region (robust) with finite-difference Jacobian, or
- Broyden (quasi-Newton) for fewer trajectory evaluations.

Scale residuals:

$$\tilde{F}_1 = \frac{F_1}{r_{\text{target}}}, \quad \tilde{F}_2 = \frac{F_2}{v_{\text{circ}}(r_{\text{target}})}.$$

16 Final “Start Coding” Input Set (Official + Explicit Assumptions)

16.1 Official inputs (from SpaceX)

- Falcon 9 diameter $d = 3.7$ m and gross mass 549 054 kg [2].
- Stage thrust: Stage 1 sea-level stage thrust 7686 kN and Stage 2 vacuum thrust 981 kN [1].
- Propellant: LOX/RP-1 for both stages [1].

16.2 Atmosphere (official standard)

- USSA76: $\rho(h)$ and $a_s(h)$ from NASA USSA76 [3], implemented via `ussa1976` [4, 5].

16.3 Assumptions you must choose (because SpaceX does not publish them here)

These are *required* to numerically integrate (28) and to time staging by mass depletion:

- $I_{sp,1}$, $I_{sp,2}$ (use constant per stage in v1).
- Stage dry masses and propellant masses (or, equivalently, $m_{1,\text{end}}$ and m_{cut} bounds).
- Drag model $C_D(M)$ (we recommend a smooth piecewise function; tune later).

Minimal viable (to run immediately):

- Choose $I_{sp,1}$ and $I_{sp,2}$ as constants; choose stage mass split consistent with total gross mass.
- Choose $C_D(M)$ simple and stable (piecewise + smoothing), and accept it as an uncertainty knob.

17 Falcon 9–Aligned Calibration (Closing the Model Without Guessing)

17.1 What is unknown and why calibration is necessary

SpaceX public sources provide (reliably) vehicle geometry and thrust levels, plus some timeline-type quantities (e.g., second-stage burn time on the Falcon 9 web page). However, these public sources do not fully specify a complete, internally consistent set of *stage propellant masses*, *stage dry masses*, and *I_{sp} curves* required to integrate the variable-mass ODE system with staging [2, 1].

Therefore, APOGEE closes the system using a **calibration problem**: we choose the *smallest set* of missing constants and solve for them using (i) exact physical identities and (ii) known Falcon 9 flight-consistent constraints. This is the standard engineering approach for early-phase ascent simulation when proprietary vehicle databases are unavailable.

17.2 Known (official) inputs used as hard constraints

We fix the following as non-negotiable inputs from SpaceX:

- Core diameter $d = 3.7$ m and gross liftoff mass $m_0 = 549\,054$ kg [2].
- Stage thrust (stage totals): Stage 1 $T_{1,\text{SL}} = 7686$ kN (sea level) and Stage 2 $T_{2,\text{vac}} = 981$ kN (vacuum) [1].
- Stage 2 burn time shown on the Falcon 9 vehicle page: $t_{2,\text{burn}} = 397$ s [2].
- Propellant type: LOX/RP-1 for both stages [1].
- Atmosphere: U.S. Standard Atmosphere 1976 (USSA76), implemented via `ussa1976`, which explicitly implements NASA TM-X-74335 [3, 4].

17.3 State and equations used during calibration

We calibrate using the same implementation-ready ODEs (Eqs. (24)–(28) in the main document):

$$\dot{r} = v \sin \gamma, \quad \dot{\lambda} = \frac{v \cos \gamma}{r}, \quad \dot{v} = \frac{T}{m} \cos \alpha - \frac{D}{m} - \frac{\mu}{r^2} \sin \gamma, \quad \dot{\gamma} = \frac{T}{mv} \sin \alpha + \left(\frac{v}{r} - \frac{\mu}{r^2 v} \right) \cos \gamma, \quad \dot{m} = -\frac{T}{I_{sp} g_0}.$$

During gravity-turn guidance we enforce $\alpha = 0$ (thrust aligned with velocity). Drag is

$$D = \frac{1}{2} \rho(h) C_D(M) A_{\text{ref}} v^2, \quad A_{\text{ref}} = \pi \left(\frac{d}{2} \right)^2, \quad M = \frac{v}{a_s(h)}.$$

17.4 Calibration variables (minimal set)

To obtain a numerically closed model that is Falcon 9-aligned but still simple, we define the *calibration parameter vector*

$$\mathbf{p} = \begin{bmatrix} I_{sp,1} \\ I_{sp,2} \\ t_{1,\text{burn}} \\ m_{1,\text{dry}} \\ m_{2,\text{dry}} \\ \mathbf{c}_D \end{bmatrix}, \quad (35)$$

where $\mathbf{c}_D$ are the few parameters defining a smooth $C_D(M)$ model (see Sec. 17.10).

Why include $t_{1,\text{burn}}$? The Falcon 9 web page explicitly lists Stage 2 burn time [2]; Stage 1 burn time is not consistently presented with the same clarity in those same official sources. To remain strict and avoid mixing inconsistent datasets, we treat $t_{1,\text{burn}}$ as a calibration variable constrained to the physically plausible range of Stage 1 burn duration (close to the commonly cited ~ 160 s). If you later confirm a Stage 1 burn time from an official source in your docs, lock it as a constant.

17.5 Stage propellant masses implied by thrust, burn time, and I_{sp}

The variable-mass identity

$$\dot{m} = -\frac{T}{I_{sp} g_0}$$

implies that for a stage with constant (T, I_{sp}) during its burn, the propellant mass consumed is

$$m_{i,\text{prop}}(\mathbf{p}) = \int_0^{t_{i,\text{burn}}} \frac{T_i}{I_{sp,i} g_0} dt = \frac{T_i t_{i,\text{burn}}}{I_{sp,i} g_0}. \quad (36)$$

This is **not an assumption**; it follows directly from the definition of I_{sp} and mass conservation.

17.6 Mass accounting constraint (hard equality)

At liftoff, the gross mass must equal the sum of dry masses, propellant masses, and payload:

$$m_0 = m_{1,\text{dry}} + m_{2,\text{dry}} + m_{1,\text{prop}}(\mathbf{p}) + m_{2,\text{prop}}(\mathbf{p}) + m_{\text{PL}}, \quad (37)$$

where m_{PL} is **parameterized** (mission input). Equation (37) is enforced as a strict constraint in calibration.

17.7 Falcon 9 flight-consistency constraints (targets)

We do *not* require reproducing a specific mission; instead we enforce a small set of flight-consistency constraints that any Falcon 9-like ascent must satisfy:

(1) Orbit insertion conditions (hard targets). At final cutoff (SECO), for a circular orbit at $r_{\text{target}} = R_E + h_{\text{target}}$:

$$r_{\text{SECO}} = r_{\text{target}}, \quad v_{\text{SECO}} = \sqrt{\frac{\mu}{r_{\text{target}}}}, \quad \gamma_{\text{SECO}} \approx 0. \quad (38)$$

(We typically enforce the first two as hard equalities via shooting; the third as a small tolerance.)

(2) Stage event ordering (hard logic). Stage 1 burn $\rightarrow$ MECO $\rightarrow$ separation $\rightarrow$ Stage 2 ignition $\rightarrow$ SECO. This is implemented via event functions on $m(t)$ and time.

(3) Dynamic pressure sanity (soft constraint). Maximum dynamic pressure

$$q(t) = \frac{1}{2} \rho(h(t)) v(t)^2$$

must occur in the lower atmosphere and remain within a plausible range. Since SpaceX throttle schedules are not modeled, we treat this as a diagnostic or a soft penalty to prevent unrealistic drag settings. (We do *not* force a specific Max-Q value unless you decide to.)

17.8 Objective function (strict, dimensionless, well-conditioned)

Define residuals for a single simulation run:

$$F_1(\mathbf{p}, \mathbf{u}) = \frac{r_{\text{SECO}}(\mathbf{p}, \mathbf{u}) - r_{\text{target}}}{r_{\text{target}}}, \quad (39)$$

$$F_2(\mathbf{p}, \mathbf{u}) = \frac{v_{\text{SECO}}(\mathbf{p}, \mathbf{u}) - \sqrt{\mu/r_{\text{target}}}}{\sqrt{\mu/r_{\text{target}}}}, \quad (40)$$

$$F_3(\mathbf{p}, \mathbf{u}) = \frac{\gamma_{\text{SECO}}(\mathbf{p}, \mathbf{u})}{\gamma_{\text{scale}}}, \quad (41)$$

$$F_4(\mathbf{p}) = \frac{m_0 - (m_{1,\text{dry}} + m_{2,\text{dry}} + m_{1,\text{prop}}(\mathbf{p}) + m_{2,\text{prop}}(\mathbf{p}) + m_{\text{PL}})}{m_0}, \quad (42)$$

where γ_{scale} is a small scaling (e.g., $\gamma_{\text{scale}} = 1^\circ$ in radians).

We split unknowns into:

- **Guidance/shooting variables $\mathbf{u}$** (solved per mission): $\mathbf{u} = [\theta_0, m_{\text{cut}}]^T$.
- **Vehicle calibration parameters $\mathbf{p}$** (solved once, reused).

The calibration solves:

$$\min_{\mathbf{p}} \|\mathbf{F}(\mathbf{p}, \mathbf{u}^*(\mathbf{p}))\|_2^2 + \beta \Phi(\mathbf{p}), \quad (43)$$

subject to physical bounds (positivity, plausible I_{sp} ranges, etc.), where:

- $\mathbf{u}^*(\mathbf{p})$ is obtained by an **inner shooting solve** enforcing (38) for a reference mission (e.g., $h_{\text{target}} = 200$ km, chosen once).

- $\Phi(\mathbf{p})$ is a **regularizer** encoding “do not choose absurd values”, e.g.,

$$\Phi(\mathbf{p}) = \left(\frac{I_{sp,1} - \bar{I}_{sp,1}}{\sigma_{I1}} \right)^2 + \left(\frac{I_{sp,2} - \bar{I}_{sp,2}}{\sigma_{I2}} \right)^2 + \left(\frac{t_{1,\text{burn}} - \bar{t}_1}{\sigma_{t1}} \right)^2,$$

with $\bar{I}_{sp,i}$ and $\bar{t}_1$ chosen as nominal engineering values.

Strictness note. Equation (36) and mass budget (37) ensure full dimensional consistency and remove degrees of freedom. The regularizer Φ is not “cheating”: it is the formal way to encode prior engineering knowledge when public data is incomplete.

17.9 Recommended numerical solution strategy (robust)

17.9.1 Inner loop: mission shooting

Given $\mathbf{p}$, solve for $\mathbf{u} = [\theta_0, m_{\text{cut}}]$ by enforcing two equalities:

$$r_{\text{SECO}}(\mathbf{p}, \mathbf{u}) - r_{\text{target}} = 0, \quad v_{\text{SECO}}(\mathbf{p}, \mathbf{u}) - \sqrt{\mu/r_{\text{target}}} = 0.$$

Use a robust root-finder (Powell hybrid / trust-region) with finite-difference Jacobian. Each residual evaluation integrates the ODEs with event handling.

17.9.2 Outer loop: calibration

With $\mathbf{u}^*(\mathbf{p})$ from the inner loop, minimize (43) using a derivative-free trust-region method (or quasi-Newton if you later move to autodiff). Because each function evaluation is expensive (full trajectory), keep $\mathbf{p}$ low-dimensional.

17.9.3 ODE integration

Use adaptive embedded Runge–Kutta 5(4) (Tsit5 or Dopri5) with event detection. This is appropriate because the system is nonlinear but not stiff in typical ascent regimes.

17.10 Aerodynamic model to calibrate (simple, stable)

SpaceX does not publish Falcon 9 $C_D(M)$; thus we use a smooth function with few parameters:

$$C_D(M) = C_{D,\text{sub}} + \Delta C_{D,\text{tr}} S(M; M_1, M_2) + \Delta C_{D,\text{sup}} S(M; M_3, M_4), \quad (44)$$

where S is a smooth step (e.g., logistic or tanh-based), and $(M_1 < M_2 < M_3 < M_4)$ define transition ranges (subsonic→transonic→supersonic). The parameter vector $\mathbf{c}_D$ consists of these amplitudes and transition Mach numbers. This ensures differentiability and numerical stability.

17.11 Initial conditions and “Ecuador launch” (with no Earth rotation)

With Earth rotation explicitly neglected, the only consistent interpretation of “Ecuador launch” is that the trajectory is modeled in the equatorial plane and the target orbit is equatorial. The initial condition is:

$$r(0) = R_E, \quad \lambda(0) = 0, \quad v(0) = 0, \quad \gamma(0) = \frac{\pi}{2}, \quad m(0) = m_0.$$

No extra inertial boost is applied.

17.12 Calibration deliverable (what you hand to coding AIs)

The calibration outputs a fully specified parameter set:

$$\{I_{sp,1}, I_{sp,2}, t_{1,burn}, m_{1,dry}, m_{2,dry}, C_D(M) \text{ parameters}\}$$

together with the fixed official constants ($d, m_0, T_{1,SL}, T_{2,vac}, t_{2,burn}$) and atmosphere model choice (USSA76 via `ussa1976`).

17.13 Key strict validation checks (must pass)

Any calibrated solution is rejected unless:

- Mass budget (37) holds to numerical tolerance ($< 10^{-8}$ relative).
- $m_{i,prop} > 0, m_{i,dry} > 0$.
- Orbit diagnostics satisfy near-circularity at insertion: $|e| < e_{\max}$ (e.g. 10^{-3}) and $|r_{apo} - r_{peri}|$ small compared to r_{target} .
- Dynamic pressure $q(t)$ remains physically plausible and peaks in the lower atmosphere (diagnostic for C_D sanity).

18 IEEE References

References

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